Facilities and Other Resources

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PI laboratory, BYU Department of Mechanical Engineering. A 700 square foot powder metallurgy laboratory in the Clyde Building (CB 154), assigned to the PI, holds an AMAZEMET rePOWDER ultrasonic atomizer with an induction crucible module. We use the atomizer to make alloy powder from elemental and master-alloy feedstock weighed into machined cups, about four compositions per week and about 0.5 kg per run. For this project the cups are oxygen-free copper and the powders are copper alloys. They are melted at up to 1,300 °C in graphite crucibles and atomized onto coated carbon-carbon plates. For melts that contain nickel or iron, the crucible is sprayed with boron nitride and the nozzle is boron nitride. A powder doser built in the laboratory, an open-source auger doser that weighs each dose on an analytical balance (0.1 mg), dispenses the powdered minor additions into each charge and logs every dose. Sieves (ASTM B214), a Hall flowmeter funnel (ASTM B212 and B213), glove bags and dry powder storage support the powder checks. The room has a canopy hood, a fume hood, an industrial dehumidifier, and electrical, gas, compressed air and chilled water services installed with BYU Environmental Health and Safety. It connects directly to the BYU Microscopy Center by a door with no public access.

BYU Microscopy Center. Scanning electron microscopes with energy-dispersive X-ray spectroscopy (EDS) and electron backscatter diffraction (EBSD) (Thermo Fisher Apreo C, Helios 600 with a focused ion beam, and Verios), and a Tecnai TF-20 transmission electron microscope. The instruments are shared, have high availability, and are charged at internal hourly rates (in the budget).

Other BYU facilities. Metallographic preparation uses PX500 automatic and PX300 manual polishers, vibratory polishers, an etching fume hood and an electropolisher, with a Keyence VHX-7000 digital optical microscope for inspection. We measure composition and structure at the BYU X-ray Diffraction Facility and the BYU Spectroscopic Facility (Agilent 8900 ICP-MS/MS and XPS). The BYU Prototyping Lab machines charge cups and test specimens by CNC. Sub-size tensile tests (ASTM E8/E8M) run on the Instron load frames of the materials testing laboratory in the Department of Mechanical Engineering [TODO Sterling: confirm]. We measure particle size distributions by sieve analysis in the PI laboratory and by image analysis of SEM micrographs.

Computing. The BYU Office of Research Computing operates a shared cluster of about 28,000 CPU cores and 367 GPUs on 616 nodes (`https://rc.byu.edu`, 2026), shared among BYU researchers at no direct cost. CALPHAD calculations use Thermo-Calc with its copper alloy database [TODO Sterling: confirm the BYU license includes it] and the open-source pycalphad; heat-flow calculations and the search software run on the same cluster.

Directed energy deposition. Single-stream, blown-powder laser DED is not available at BYU. The project buys machine time at two facilities, and the search treats them as two steps of different cost. Neither facility is a subawardee. *The small step* is the InssTek MX-Lab in the laboratory of Professor Z. Zak Fang (Department of Metallurgical Engineering, University of Utah, Salt Lake City, 45 miles from BYU), which his group has offered to the project. It has a 500 W fiber laser, 150 mm of travel on each axis, control of melt-pool height, and six powder feeders that can be run separately. It makes pre-placed tracks from a few grams of powder and blown-powder tracks,

thin walls and small blocks from about 0.5 kg, on a day trip from BYU. Its use is budgeted as hourly instrument time until the group sets a rate [TODO Sterling: confirm with Fang: the rate, the smallest powder charge per feeder, the argon chamber and any preheat, and who runs it]. *The large step* is the FormAlloy X2 blown-powder laser DED system at the Colorado School of Mines (ADAPT center and Shared Instrumentation Facility, Golden, Colorado), bought on a purchase order. Facility staff run it at posted external rates for machine time and labor (in the budget). A typical session of four 25 mm cubes takes about 3 machine hours and 2 labor hours. The X2 takes powder from 50 to 200 μm, needs about 2 kg of each powder to fill its hoppers, catches about half of the powder fed in the melt pool, and has deposited copper-nickel tracks. It builds the tensile and corrosion blocks and the repair coupons on thick plate, which need more laser power than the MX-Lab has [TODO Sterling: the X2's laser]. The United States representative of InssTek has also offered deposition within the InssTek network, which lists nickel-aluminum bronze (UNS C95300) among its qualified materials.

Corrosion testing. The electrochemical screen runs on the liquid-handling gantry in the PI laboratory (CubXL, Ursa Laboratories: a pipette, a vial capper and two cameras on a scripted gantry with about 390 $\times$ 230 $\times$ 120 mm of travel). The gantry positions small O-ring cells on polished specimens, with the potentiostat requested in the Equipment section. Immersion tests in substitute ocean water (ASTM D1141) run in capped vials on the deck of the gantry, which samples the solutions for ICP-MS at the BYU Spectroscopic Facility. Oxygen and carbon analysis of powder lots is purchased from a commercial laboratory (inert gas fusion for oxygen and combustion for carbon).

Government-owned facilities. None are required. No additional facilities will be acquired at the applicant's expense.